

Chapter 5: Expected Utility Theory

ECON 5001

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Based on: *Game Theory* by Giacomo Bonanno

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Beyond Risk Neutrality

Chapter 4: chance moves give us **lotteries** over outcomes

Our shortcut was to assume:

- All outcomes are money
- Players are **risk neutral** (rank lotteries by expected value)

That leaves out a lot:

- People buy insurance. (And lottery tickets!)
- Many outcomes aren't money at all
 - Win/lose/draw, getting the job, where to eat dinner...

Our goal:

1. Rank lotteries over **any** kind of outcome
2. Describe different attitudes toward risk

MobLab: “Bomb Risk Game”

- MobLab time!
- There are 100 boxes. One of them hides a **bomb**
 - Every box is equally likely to have it
- You choose how many boxes to open (anything from 0 to 100)
- Then we find out where the bomb was:
 - **Not** in your boxes: you earn 1 point per box you opened
 - **In** your boxes: you earn 0

Feel free to talk! And play it the way you'd actually play it.

Lotteries

Lotteries

A **lottery** L is a list of outcomes and their probabilities:

$$L = \begin{pmatrix} o_1 & o_2 & \cdots & o_n \\ p_1 & p_2 & \cdots & p_n \end{pmatrix}$$

The outcomes can be **anything**:

- Money: $L_1 = \begin{pmatrix} \$10 & \$20 \\ \frac{1}{2} & \frac{1}{2} \end{pmatrix}$

- Not money: $L_2 = \begin{pmatrix} \text{Buckeyes beat Michigan} & \text{Buckeyes lose to Michigan} \\ \frac{3}{5} & \frac{2}{5} \end{pmatrix}$

A sure thing is a lottery too. Just write o for “outcome o with probability 1”

Expected Value: Numbers Only

Recall the **expected value** of a money lottery:

$$E[L] = p_1x_1 + p_2x_2 + \cdots + p_nx_n$$

$$L_1 = \begin{pmatrix} \$10 & \$20 \\ \frac{1}{2} & \frac{1}{2} \end{pmatrix} \quad E[L_1] = \underline{\hspace{2cm}}$$

$$L_2 = \begin{pmatrix} \text{Beat Michigan} & \text{Lose to Michigan} \\ \frac{3}{5} & \frac{2}{5} \end{pmatrix} \quad E[L_2] =$$

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Lesson: We can calculate an expected value when outcomes are **numbers** (like money).
When outcomes are **not** numbers, expected value doesn't exist!

So how do we rank lotteries like L_2 ?

Expected Utility

Two Kinds of Utility

Step 1: Give every *outcome* a number

- $u(o)$ is the **utility index** of outcome o (small u : outcomes)

Step 2: Use those to give every *lottery* a number

- $U(L)$ is the **expected utility** of lottery L (capital U : lotteries)

Expected utility: average the utility index, weighting by the probabilities.

$$\underbrace{U(L)}_{\text{expected utility of the lottery}} = \underbrace{p_1 u(o_1) + p_2 u(o_2) + \cdots + p_n u(o_n)}_{\text{probability-weighted average of the utility index}}$$

Bonanno writes $U(o)$ for our $u(o)$, and $E[U(L)]$ for our $U(L)$.

Example

Alex's utility index over dinner outcomes:

Outcome o	Cazuela's	Yoshi Sushi	Dining hall
Utility index $u(o)$	10	6	0

Two lotteries:

$$L_A = \begin{pmatrix} \text{Yoshi Sushi} \\ 1 \end{pmatrix} \qquad L_B = \begin{pmatrix} \text{Cazuela's} & \text{Dining hall} \\ \frac{1}{2} & \frac{1}{2} \end{pmatrix}$$

$$U(L_A) =$$

$$U(L_B) =$$

Alex prefers: _____

Note: no expected *value* here. But expected *utility* works fine.

Why Expected Utility?

Do people really rank lotteries by expected utility?

Theorem (von Neumann & Morgenstern, 1944)

If your ranking of lotteries satisfies a few “rationality” axioms, then there is a utility index $u(o)$ such that

$$L \succ L' \iff U(L) > U(L') \quad \text{and} \quad L \sim L' \iff U(L) = U(L')$$

In words: you act *as if* you’re maximizing expected utility

The axioms are in Bonanno §5.3. We’re going to skip them.

- Interested? Read through them for fun!

From here on: we **assume** players maximize expected utility



John von Neumann (1903–1957)



Oskar Morgenstern (1902–1977)

- **von Neumann:** Budapest math prodigy. Princeton IAS (with Einstein)
 - Manhattan Project, modern computer design, minimax theorem
- **Morgenstern:** Vienna economist. Left Austria in 1938
- **1944:** *Theory of Games and Economic Behavior*. Game theory is born
- Expected utility axioms: an appendix to the 1947 edition!

Rescaling the Utility Index

Can We Rescale Any Way We Want?

Chapter 2: utility only *ranked* outcomes. Any increasing rescaling was fine.

Is that still true? Try squaring Alex's utility index: $v(o) = u(o)^2$

Outcome o	Cazuela's	Yoshi Sushi	Dining hall
$u(o)$	10	6	0
$v(o) = u(o)^2$	100	36	0

Using v : $V(L_A) =$

$V(L_B) =$

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Using v : $V(L_A) =$ $V(L_B) =$

Same ranking of *outcomes*. Different ranking of *lotteries*!!

Lesson: The utility index can **not** be rescaled arbitrarily

Linear Rescaling

Which rescalings *are* allowed? Only **linear** ones:

$$v(o) = a \cdot u(o) + b \quad (\text{with } a > 0)$$

- You can **multiply** by a positive number a
- You can **add** (or subtract) a constant b

Try $v(o) = 2u(o) + 3$ on Alex's index (10, 6, 0):

$$v = (\quad) \quad V(L_A) = \quad \quad V(L_B) =$$

Theorem: u and v rank all lotteries the same way $\iff v$ is a linear rescaling of u

Like temperature: $^{\circ}\text{F} = 1.8 \times ^{\circ}\text{C} + 32$. Different numbers, same information.

Bonanno calls a linear rescaling a “positive affine transformation.”

Normalizing

Since linear rescaling is allowed, we can always pick a convenient scale:

$$u(\text{best outcome}) = 1$$

$$u(\text{worst outcome}) = 0$$

This is called the **normalized** utility index. How to get it:

1. Subtract $u(\text{worst})$ from every number (now the worst is 0)
2. Divide every number by the new $u(\text{best})$ (now the best is 1)

Outcome	o_1	o_2	o_3	o_4
$u(o)$	8	2	-2	0
After step 1				
After step 2				

Attitudes Toward Risk

Risk Averse, Risk Neutral, Risk Loving

Back to **money** lotteries, where $E[L]$ exists. Offer someone a choice:

the lottery L vs. getting $\$E[L]$ for sure

Definition

- **Risk averse:** prefers $\$E[L]$ for sure
- **Risk neutral:** indifferent between them
- **Risk loving:** prefers the lottery L

Ex: $L_1 = \begin{pmatrix} \$10 & \$20 \\ \frac{1}{2} & \frac{1}{2} \end{pmatrix}$ vs. $\$15$ for sure. Which would you take?

Important: This only makes sense when outcomes are **numbers** (like money).
No $E[L]$ means no “risk averse” or “risk loving.” (Bonanno doesn’t say this.)

Risk Attitudes and the Shape of $u(o)$

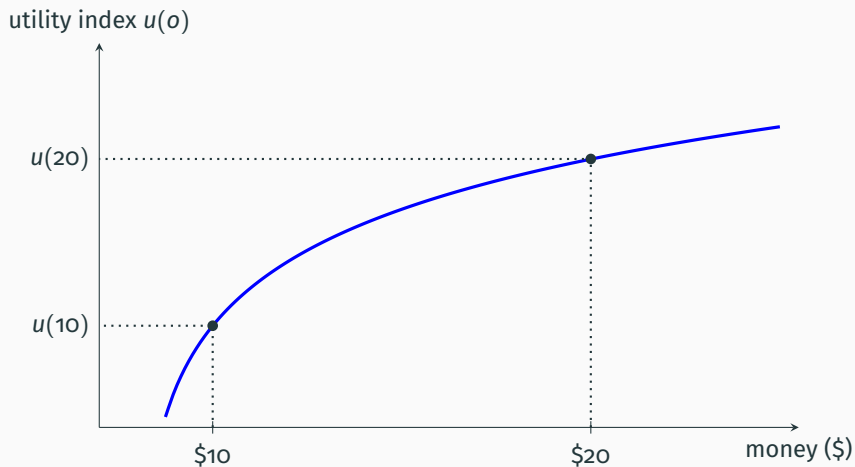
With expected utility, the definition becomes a comparison of two numbers:

$$\underbrace{U(L)}_{\text{expected utility of the lottery}} \quad \text{vs.} \quad \underbrace{u(E[L])}_{\text{utility of the expected value}}$$

Risk averse	$\iff$	$U(L) < u(E[L])$	$\iff$	$u(o)$ is concave
Risk neutral	$\iff$	$U(L) = u(E[L])$	$\iff$	$u(o)$ is linear
Risk loving	$\iff$	$U(L) > u(E[L])$	$\iff$	$u(o)$ is convex

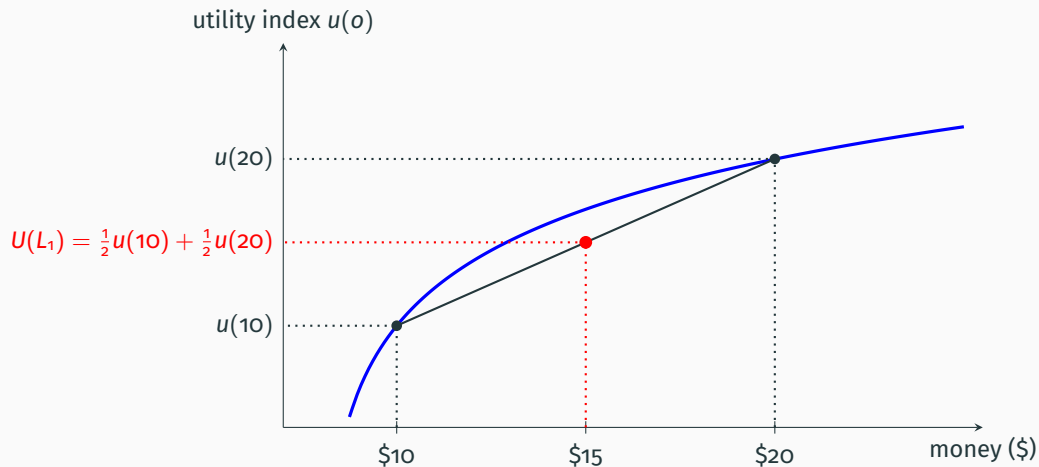
This isn't in Bonanno, but it's how economists usually think about risk attitudes.

Risk Averse $\iff$ Concave



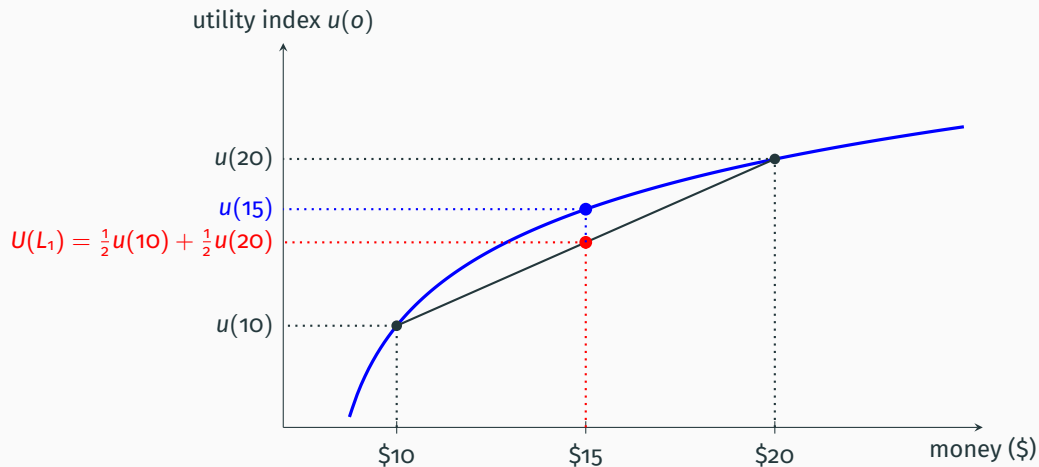
L_1 pays $\$10$ or $\$20$ with equal chance. $E[L_1] = \$15$.

Risk Averse $\iff$ Concave



$U(L_1)$ is halfway up between $u(10)$ and $u(20)$: the midpoint of the straight line.

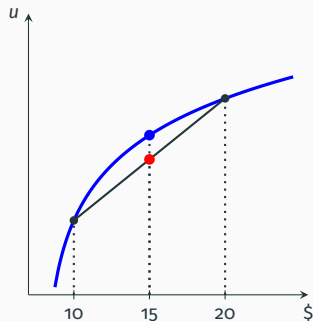
Risk Averse $\iff$ Concave



The curve is *above* the line, so $u(15) > U(L_1)$. They'd rather have \$15 for sure.

All Three Cases

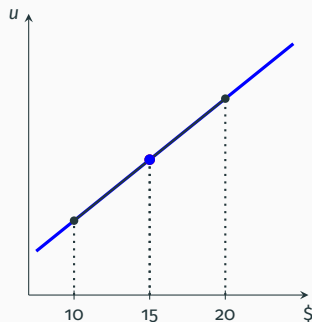
Concave



$$U(L_1) < u(15)$$

Risk averse

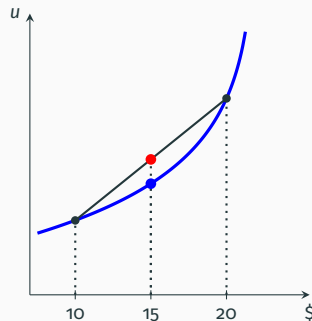
Linear



$$U(L_1) = u(15)$$

Risk neutral

Convex



$$U(L_1) > u(15)$$

Risk loving

Red dot: expected utility of the lottery. **Blue dot:** utility of $\$15$ for sure.

Back to the Bomb Game

Open k boxes: you earn k with probability $\frac{100-k}{100}$, otherwise 0

Expected value: $E[L_k] = k \cdot \frac{100 - k}{100}$

- Biggest when $k =$ _____
- Opening **exactly** that many: risk _____
- Opening **fewer**: risk _____
- Opening **more**: risk _____

Our class median: _____

Lab result: the typical person opens a bit fewer than 50

Source: Crosetto & Filippin (2013), *Journal of Risk and Uncertainty*.

Risk Attitudes in the Real World

Lots of real decisions boil down to “lottery vs. sure thing”:

- **Insurance.** Pay a premium for sure, or risk a big loss?
- **Extended warranties.** Same thing, for your laptop
- **Salary vs. commission.** Steady paycheck, or pay that depends on sales?
- **Index funds vs. single stocks.** Spread the risk, or bet on one company?
- **Sports betting, casinos, Ohio Lottery.** Pay to *take on* risk!

Real versions are messier:

- The same person might buy insurance *and* lottery tickets
- People treat gains and losses differently

But the core idea is the same: **the shape of $u(o)$ tells you how they handle risk**

Example: Car Insurance

Should You Insure Your Car?

Ruth and Logan are OSU students. Each of them has:

- \$1,000 in the bank, and a car worth \$15,000 (total wealth: \$16,000)
- A 25% chance of totaling the car this year

Nationwide offers **full insurance** for a \$6,000 premium:

- If the car is totaled, they pay you the full \$15,000

Final wealth (in \$1,000s):

	Car is fine (75%)	Car is totaled (25%)
No insurance	16	1
Insurance	10	10

$E[\text{No insurance}] =$ _____

A risk-neutral person would choose: _____

Two Different Utility Indices

Let o be final wealth (in \$1,000s).

Ruth: $u_R(o) = \sqrt{o}$

$$U_R(\text{No ins.}) = \frac{3}{4}\sqrt{16} + \frac{1}{4}\sqrt{1}$$

$$u_R(10) = \sqrt{10} \approx 3.16$$

Ruth chooses: _____

Handy: $\ln(16) \approx 2.77$ and $\ln(1) = 0$.

Logan: $u_L(o) = \ln(o)$

$$U_L(\text{No ins.}) = \frac{3}{4}\ln(16) + \frac{1}{4}\ln(1)$$

$$u_L(10) = \ln(10) \approx 2.30$$

Logan chooses: _____

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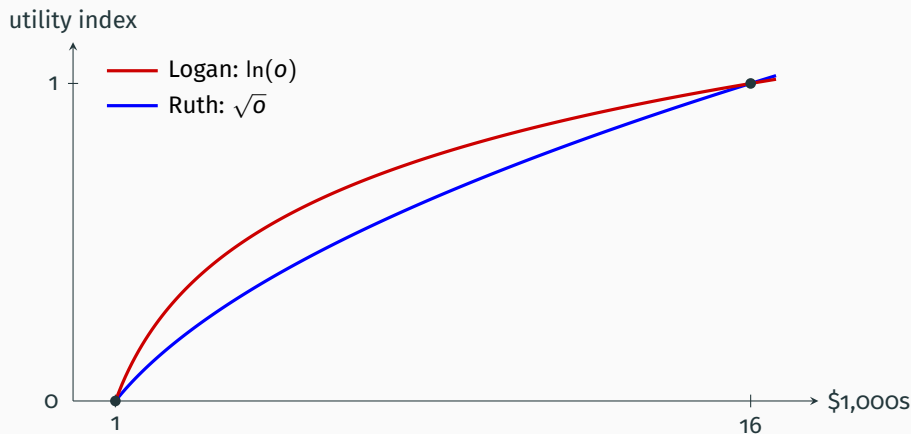
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$$u_L(10) = \ln(10) \approx 2.30$$

Logan chooses: _____

Both are concave, so **both** are risk averse. But only one buys insurance!

Seeing It in a Graph

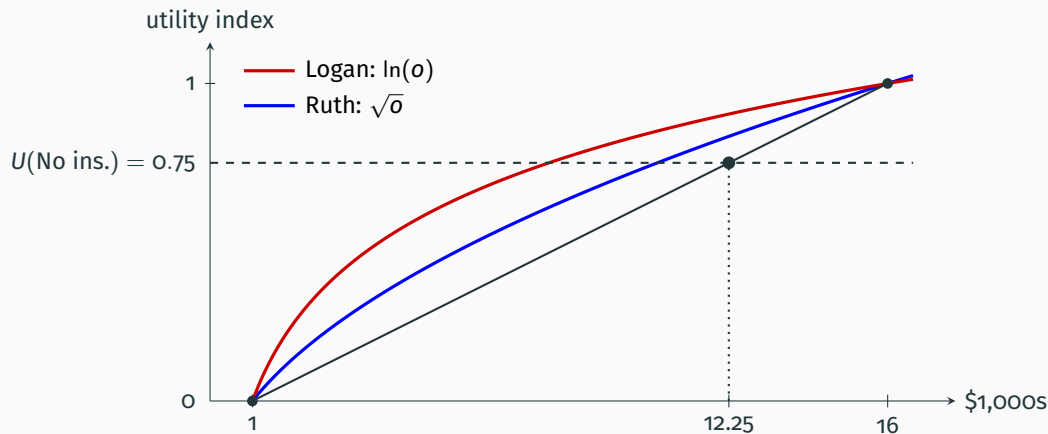


Both rescaled so that $u(1) = 0$ and $u(16) = 1$. (Linear rescaling: allowed!)

Logan's curve bends more: Logan is **more** risk averse than Ruth.

Normalized: Ruth $u_R(o) = \frac{1}{5}\sqrt{o} - \frac{1}{5}$ Logan $u_L(o) = \frac{1}{5\ln(4)}\ln(o) \approx 0.36\ln(o)$

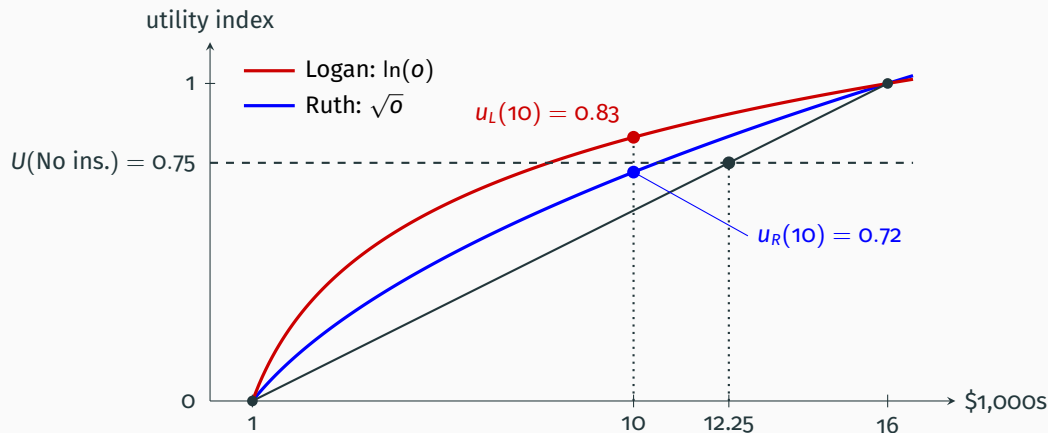
Seeing It in a Graph



No insurance: 75% chance of the best, 25% chance of the worst.
So $U(\text{No ins.}) = 0.75$ for **both** of them: the dashed line.

Normalized: Ruth $u_R(o) = \frac{1}{2}\sqrt{o} - \frac{1}{2}$ Logan $u_L(o) = \frac{1}{\ln(16)} \ln(o) \approx 0.36 \ln(o)$

Seeing It in a Graph



Insurance gives 10 for sure. Is $u(10)$ above or below the dashed line?

Ruth: below, so no insurance. **Logan:** above, so buy insurance.

Normalized: Ruth $u_R(o) = \frac{1}{2}\sqrt{o} - \frac{1}{2}$ Logan $u_L(o) = \frac{1}{\ln(16)} \ln(o) \approx 0.36 \ln(o)$

How Much Would They Pay?

The expected loss is $\frac{1}{4} \times \$15,000 = \$3,750$. Nationwide charges \$6,000.

What's the **most** each would pay for full insurance?

- **Ruth:** find the sure wealth w with $\sqrt{w} = 3.25$
- **Logan:** find the sure wealth w with $\ln(w) = \frac{3}{4} \ln(16)$

On the graph: where the dashed line crosses each curve.

Lesson: Knowing someone is risk averse doesn't tell you what they'll choose. You need to know their utility index.

Let's Work One Out

The Prize Wheel

Carmen gets one spin of a prize wheel at the Ohio Union. The possible prizes:

- M : a ticket to the Michigan game J : a pint of Jeni's ice cream
- D : a dozen Buckeye Donuts N : nothing

Carmen's ranking: $M \succ J \succ D \succ N$. Carmen also tells us:

- J for sure is just as good as a 50% chance of M (otherwise N)
- D for sure is just as good as a 20% chance of M (otherwise N)

Carmen can pick which wheel to spin:

$$W_1 = \begin{pmatrix} J & D \\ 0.5 & 0.5 \end{pmatrix} \quad W_2 = \begin{pmatrix} M & N \\ 0.3 & 0.7 \end{pmatrix} \quad W_3 = \begin{pmatrix} M & J & D \\ 0.1 & 0.4 & 0.5 \end{pmatrix}$$

The Prize Wheel

1. Write down Carmen's normalized utility index.
2. Find the expected utility of each wheel. Which does Carmen spin?
3. Rescale using $v(o) = 10u(o) + 5$. Does the ranking of wheels change?
4. Now rescale using $w(o) = u(o)^2$. Does the ranking change? Why?
5. Is Carmen risk averse, risk neutral, or risk loving?

What Have We Learned, Charlie Brown?

- **Expected value** needs numeric outcomes. **Expected utility** doesn't
- $u(o)$: **utility index** over outcomes
- $U(L)$: **expected utility** over lotteries
 - $U(L) = p_1 u(o_1) + \dots + p_n u(o_n)$
- The utility index can only be rescaled **linearly**
 - So we can normalize: best outcome = 1, worst outcome = 0
- Risk averse / neutral / loving $\iff u(o)$ is concave / linear / convex
 - Only for numeric outcomes!
- Two risk-averse people can still make different choices

Next: games where the payoffs are expected utilities

- Bonanno, *Game Theory*, Chapter 5
 - §5.1 (risk attitudes) and §5.2 (expected utility)
 - §5.3 (the axioms): optional, read it for fun
- Remember: Bonanno's $U(o)$ is our $u(o)$. His $E[U(L)]$ is our $U(L)$
- Exercises: §5.4.1 (risk attitudes) and §5.4.2 (expected utility)
 - Exercises 5.2 and 5.4: outcomes that aren't numbers
 - Exercise 5.11: two risk-averse people who choose differently
- Solutions in §5.5